

Problem 1

Model a mobile robot as a moving frame $\{r\}$ with $\hat{y}_r$ pointing forward and $\hat{x}_r$ pointing to the robot's right. All rotations are about $\hat{z}$, positive counter-clockwise. The robot starts with $\{r\}$ coincident with the world frame $\{0\}$. It then executes, in order: translate forward 1 m; rotate 45° clockwise; translate forward 0.5 m; rotate 30° counter-clockwise; translate backward 2 m. Denote the resulting pose $\{\text{final}\}$. From $\{\text{final}\}$ the robot observes an object 1.5 m to its right; the object frame $\{\text{obj}\}$ has the same orientation as $\{\text{final}\}$.

Part (a): Scaled Path Drawing

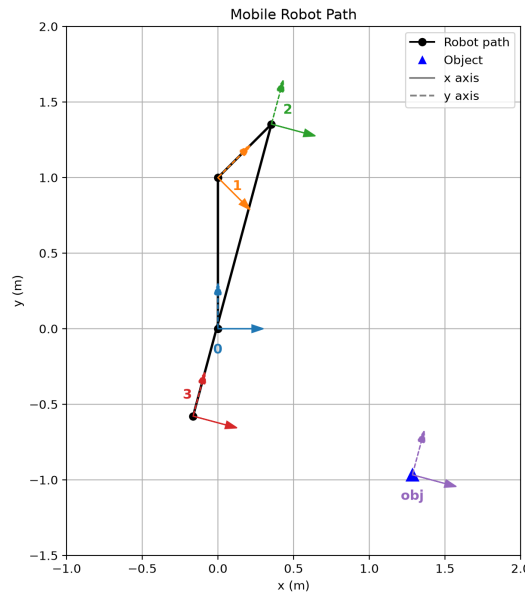


Figure 1: Scaled path of the robot frame $\{r\}$ through frames $\{0\}, \{1\}, \{2\}, \{3\} = \{\text{final}\}$, and the object frame $\{\text{obj}\}$.

Part (b): ${}^0T_{\text{final}}$ as an Ordered Product of $\text{Trans}(\cdot)$ and $\text{Rot}_z(\cdot)$

Each motion is applied in the moving (body) frame, so the transforms are post-multiplied in the order executed:

$${}^0T_{\text{final}} = \text{Trans}\begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{Rot}_z(-45^\circ) \text{Trans}\begin{pmatrix} 0 \\ 0.5 \end{pmatrix} \text{Rot}_z(30^\circ) \text{Trans}\begin{pmatrix} 0 \\ -2 \end{pmatrix}$$

Part (c): ${}^0T_{\text{final}}$

Writing the product as ${}^0T_{\text{final}} = {}^0T_1 {}^1T_2 {}^2T_3$:

$$\underbrace{\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 1 \\ 0 & 0 & 1 \end{bmatrix}}_{{}^0T_1} \underbrace{\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0.5 \\ 0 & 0 & 1 \end{bmatrix}}_{{}^1T_2} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}}_{{}^2T_3}$$

$${}^0T_{\text{final}} = \begin{bmatrix} 0.9659 & 0.2588 & -0.1641 \\ -0.2588 & 0.9659 & -0.5782 \\ 0 & 0 & 1 \end{bmatrix}$$

Part (d): ${}^{\text{final}}T_{\text{obj}}$

General form:

$${}^F T_{\text{obj}} = \begin{bmatrix} {}^F R_{\text{obj}} & {}^F d_{F \rightarrow \text{obj}} \\ \vec{0}^T & 1 \end{bmatrix}$$

The object is 1.5 m to the robot's right (i.e. along $\hat{x}_{\text{final}}$) with the same orientation as {final}, so

$${}^F R_{\text{obj}} = \mathbf{I} \text{ and } {}^F d_{F \rightarrow \text{obj}} = \begin{bmatrix} 1.5 \\ 0 \end{bmatrix}:$$

$${}^{\text{final}}T_{\text{obj}} = \begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Part (e): ${}^{\text{obj}}T_0$

First chain to the world frame:

$${}^0T_{\text{obj}} = {}^0T_{\text{final}} \cdot {}^{\text{final}}T_{\text{obj}} = \begin{bmatrix} 0.9659 & 0.2588 & 1.2848 \\ -0.2588 & 0.9659 & -0.9665 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Then invert using } {}^0T_{\text{obj}}^{-1} = \begin{bmatrix} {}^0R_{\text{obj}}^T & -{}^0R_{\text{obj}}^T {}^0d_{0 \rightarrow \text{obj}} \\ \vec{0}^T & 1 \end{bmatrix}:$$

$${}^{\text{obj}}T_0 = \begin{bmatrix} 0.9659 & -0.2588 & -1.4910 \\ 0.2588 & 0.9659 & 0.6010 \\ 0 & 0 & 1 \end{bmatrix}$$

Problem 2

Figure 2 shows a three-link manipulator. The first joint is at the origin O_0 and rotates about ${}^0\hat{z}$ with an angle θ_0 . The first link has a length of 1 m. The second link is prismatic and extends and contracts with length a_1 , it is attached at a 45° rotation clockwise about the ${}^1\hat{z}$ direction. The third joint is again rotary and rotates relative to the second link by angle θ_2 . The third link has a length of 1 m.

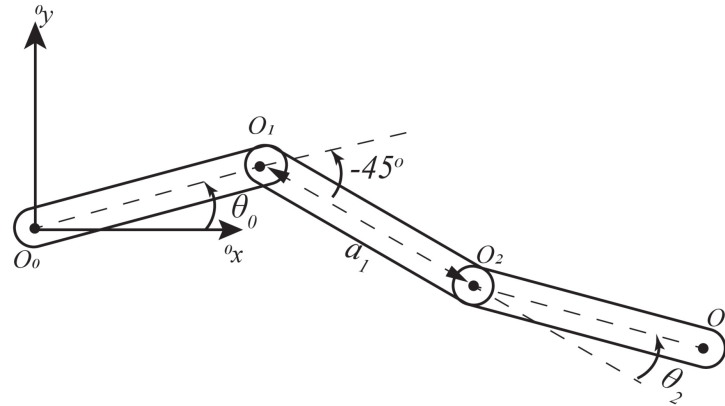


Figure 2: Three-link RPR manipulator.

Part (a): Manipulator Drawing ($\theta_0 = \pi/4$, $a_1 = 0.5 \text{ m}$, $\theta_2 = \pi/4$)

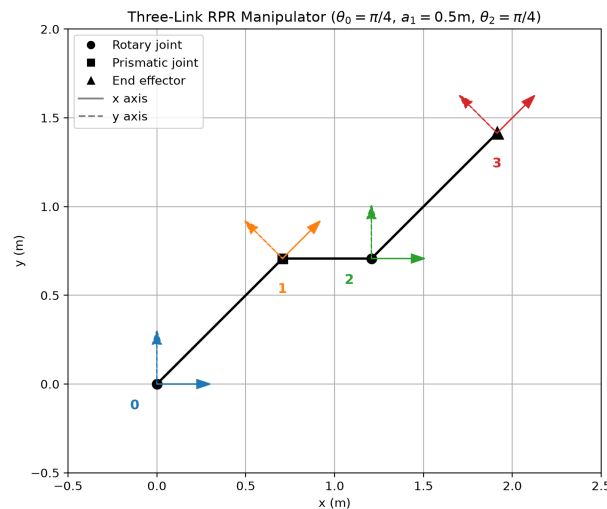


Figure 3: RPR manipulator drawn to scale for $\theta_0 = \pi/4$, $a_1 = 0.5 \text{ m}$, $\theta_2 = \pi/4$.

Part (b): 0T_1

$${}^0T_1 = \begin{bmatrix} {}^0R_1 & {}^0d_{01} \\ \vec{0}^T & 1 \end{bmatrix}, \quad {}^0R_1 = \begin{bmatrix} c\theta_0 & -s\theta_0 \\ s\theta_0 & c\theta_0 \end{bmatrix}$$

Link 1 has length 1 m along $\hat{x}_1$: ${}^1d_{01} = 1 \hat{x}_1$, so ${}^0d_{01} = 1c\theta_0 \hat{x} + 1s\theta_0 \hat{y}$.

$${}^0T_1 = \begin{bmatrix} c\theta_0 & -s\theta_0 & c\theta_0 \\ s\theta_0 & c\theta_0 & s\theta_0 \\ 0 & 0 & 1 \end{bmatrix}$$

Part (c): 1T_2

Joint 1 is attached with a fixed -45° rotation about ${}^1\hat{z}$:

$${}^1T_2 = \begin{bmatrix} {}^1R_2 & {}^1d_{12} \\ \vec{0}^T & 1 \end{bmatrix}, \quad {}^1R_2 = \begin{bmatrix} \cos(-45^\circ) & -\sin(-45^\circ) \\ \sin(-45^\circ) & \cos(-45^\circ) \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

The prismatic link has length a_1 along $\hat{x}_2$: ${}^2d_{12} = a_1 \hat{x}_2$, so ${}^1d_{12} = a_1 \frac{\sqrt{2}}{2} \hat{x} - a_1 \frac{\sqrt{2}}{2} \hat{y}$.

$${}^1T_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & a_1 \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & -a_1 \frac{\sqrt{2}}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

Part (d): 2T_3

$${}^2T_3 = \begin{bmatrix} {}^2R_3 & {}^2d_{23} \\ \vec{0}^T & 1 \end{bmatrix}, \quad {}^2R_3 = \begin{bmatrix} c\theta_2 & -s\theta_2 \\ s\theta_2 & c\theta_2 \end{bmatrix}$$

Link 3 has length 1 m along $\hat{x}_3$: ${}^3d_{23} = 1 \hat{x}_3$, so ${}^2d_{23} = c\theta_2 \hat{x} + s\theta_2 \hat{y}$.

$${}^2T_3 = \begin{bmatrix} c\theta_2 & -s\theta_2 & c\theta_2 \\ s\theta_2 & c\theta_2 & s\theta_2 \\ 0 & 0 & 1 \end{bmatrix}$$

Part (e): 0T_3

$${}^0T_3 = {}^0T_1 {}^1T_2 {}^2T_3 = \begin{bmatrix} c\theta_0 & -s\theta_0 & c\theta_0 \\ s\theta_0 & c\theta_0 & s\theta_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & a_1 \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & -a_1 \frac{\sqrt{2}}{2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta_2 & -s\theta_2 & c\theta_2 \\ s\theta_2 & c\theta_2 & s\theta_2 \\ 0 & 0 & 1 \end{bmatrix}$$

Since $\mathbf{R}(\theta_0)\mathbf{R}(-45^\circ)\mathbf{R}(\theta_2) = \mathbf{R}(\theta_0 - 45^\circ + \theta_2)$, the product simplifies to:

$${}^0\mathbf{T}_3 = \begin{bmatrix} c(\theta_0 - 45^\circ + \theta_2) & -s(\theta_0 - 45^\circ + \theta_2) & c\theta_0 + a_1c(\theta_0 - 45^\circ) + c(\theta_0 - 45^\circ + \theta_2) \\ s(\theta_0 - 45^\circ + \theta_2) & c(\theta_0 - 45^\circ + \theta_2) & s\theta_0 + a_1s(\theta_0 - 45^\circ) + s(\theta_0 - 45^\circ + \theta_2) \\ 0 & 0 & 1 \end{bmatrix}$$

Problem 3

Figure 4 shows the path of a mobile robot. The displacements are given by:

$${}^0\mathbf{d}_{01} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad {}^1\mathbf{d}_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad {}^2\mathbf{d}_{23} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad {}^3\mathbf{d}_{34} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad {}^4\mathbf{d}_{45} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

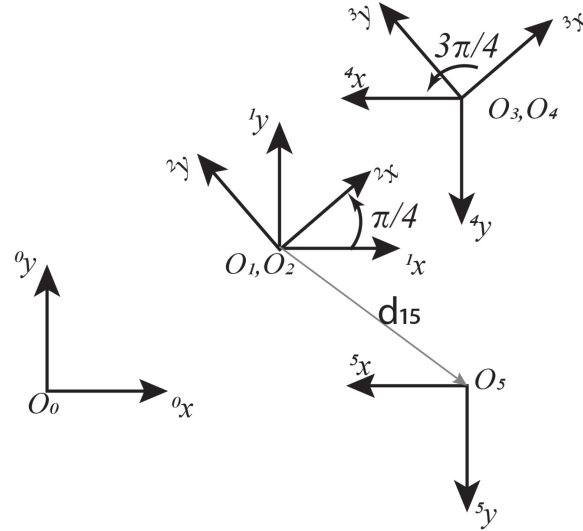


Figure 4: Path of mobile robot.

Part (a): 0T_1

$${}^0T_1 = \begin{bmatrix} {}^0R_1 & {}^0\mathbf{d}_{01} \\ \vec{0}^T & 1 \end{bmatrix}, \quad {}^0R_1 = I, \quad {}^0\mathbf{d}_{01} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$${}^0T_1 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Part (b): 1T_2

$${}^1T_2 = \begin{bmatrix} {}^1R_2 & {}^1\mathbf{d}_{12} \\ \vec{0}^T & 1 \end{bmatrix}, \quad {}^1R_2 = \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix}, \quad \theta = \pi/4, \quad {}^1\mathbf{d}_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$${}^1T_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Part (c): 2T_3

${}^2R_3 = I$ since frame $\{3\}$'s axes are parallel to frame $\{2\}$'s (Fig. 4), with ${}^2d_{23} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$:

$${}^2T_3 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Part (d): 3T_4

Frames $\{3\}$ and $\{4\}$ are coincident, related by a $3\pi/4$ rotation: $s, c(3\pi/4) = \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}$.

$${}^3T_4 = \begin{bmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Part (e): 4T_5

$${}^4T_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

Part (f): 0T_5

$${}^0T_5 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5$$

$${}^0T_5 = \begin{bmatrix} -1 & 0 & 3.41 \\ 0 & -1 & -0.59 \\ 0 & 0 & 1 \end{bmatrix}$$

Part (g): ${}^0d_{05}$

$${}^0d_{05} = \begin{bmatrix} 3.41 \\ -0.59 \end{bmatrix}$$

Part (h): 0R_5

$${}^0R_5 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Part (i): Displacement from Origin 1 to Origin 5, Expressed in Frame 5

(I) Homogeneous-transformation equation for d_{15} in frame 5. The displacement is found via ${}^5d_{51}$, obtained by mapping ${}^0d_{01}$ through ${}^0T_5^{-1}$, then negated:

$${}^0T_5^{-1} = \begin{bmatrix} {}^0R_5^{-1} & -{}^0R_5^{-1}{}^0d_{05} \\ \vec{0}^T & 1 \end{bmatrix}, \quad \begin{bmatrix} {}^5d_{51} \\ 1 \end{bmatrix} = {}^0T_5^{-1} \begin{bmatrix} {}^0d_{01} \\ 1 \end{bmatrix}, \quad {}^5d_{15} = -{}^5d_{51}$$

(II) Numerical value. Since ${}^0R_5 = -I$ satisfies $R^2 = I$ and $Rd_{05} + d_{05} = \vec{0}$, 0T_5 is an involution, so ${}^0T_5^{-1} = {}^0T_5$ numerically. The matrix below is not a copy-paste of the forward transform; it is the correct inverse:

$${}^0T_5^{-1} = \begin{bmatrix} -1 & 0 & 3.41 \\ 0 & -1 & -0.59 \\ 0 & 0 & 1 \end{bmatrix}, \quad {}^5d_{51} = {}^0T_5^{-1} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.41 \\ -1.59 \end{bmatrix}$$

$${}^5d_{15} = \begin{bmatrix} -1.41 \\ 1.59 \end{bmatrix}$$

Problem 4

Determine if the following matrices are rotations and explain why or why not.

Part (a)

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Expanding along the first row: $\det = 0 \cdot (\dots) - 1 \cdot (1 \cdot 1 - 0 \cdot 0) + 0 \cdot (\dots) = -1$.

$\det = -1 \neq 1$, so this is **not a rotation** (it is a reflection).

Part (b)

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Each column has unit norm, columns are mutually orthogonal ($\mathbf{c}_i \cdot \mathbf{c}_j = 0$), and $\det(\mathbf{B}) = 1$.

Yes, this is a rotation.

Part (c)

$$\begin{bmatrix} -0.254 & 0.087 & -0.963 \\ 0.967 & 0.0169 & -0.253 \\ -0.00572 & -0.996 & -0.0884 \end{bmatrix}$$

$\|\mathbf{c}_i\|_2 = 1$, $\mathbf{c}_i \cdot \mathbf{c}_j = 0$, and $\det(\mathbf{C}) = 1$.

Yes, this is a rotation.

Part (d)

$$\begin{bmatrix} 0.546 & 0.0719 & -1.47 \\ 0.814 & -0.696 & -0.0502 \\ -1.23 & -0.813 & 0.474 \end{bmatrix}$$

$\|\mathbf{c}_i\|_2 \neq 1$ (e.g. entries such as -1.47 and -1.23 already exceed unit magnitude).

Not a rotation.

Problem 5

Complete the below rotation matrices. How do you know it is correct?

Part (a)

$$\begin{bmatrix} 0 & 0 & ? \\ 0 & 1 & ? \\ 1 & 0 & ? \end{bmatrix}$$

For a valid rotation the columns form a right-handed orthonormal set, so $\mathbf{c}_3 = \mathbf{c}_1 \times \mathbf{c}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$.

$$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

This is correct because $\mathbf{c}_3 = \mathbf{c}_1 \times \mathbf{c}_2$ guarantees unit norm, mutual orthogonality, and $\det = +1$.

Part (b)

$$\begin{bmatrix} -1 & ? & ? \\ ? & 1 & ? \\ ? & ? & ? \end{bmatrix}$$

With $\mathbf{c}_1 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{c}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$: $\mathbf{c}_3 = \mathbf{c}_1 \times \mathbf{c}_2 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$.

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Correct by the same right-hand-rule / orthonormality argument as part (a).

Part (c)

$$\begin{bmatrix} 0.3698 & ? & 0.8665 \\ -0.9276 & ? & 0.324 \\ -0.05236 & ? & 0.3797 \end{bmatrix}$$

Here $\mathbf{c}_1, \mathbf{c}_3$ are given, so $\mathbf{c}_2 = \mathbf{c}_3 \times \mathbf{c}_1$:

$$\mathbf{c}_2 = \begin{bmatrix} 0.8665 \\ 0.324 \\ 0.3797 \end{bmatrix} \times \begin{bmatrix} 0.3698 \\ -0.9276 \\ -0.05236 \end{bmatrix} = \begin{bmatrix} 0.3352 \\ 0.1858 \\ -0.9235 \end{bmatrix}$$

$$\begin{bmatrix} 0.3698 & 0.3352 & 0.8665 \\ -0.9276 & 0.1858 & 0.324 \\ -0.05236 & -0.9235 & 0.3797 \end{bmatrix}$$

Correct since $\mathbf{c}_2 = \mathbf{c}_3 \times \mathbf{c}_1$ reproduces a right-handed orthonormal column set.

Part (d)

$$\begin{bmatrix} ? & 0.1166 & ? \\ -0.2039 & ? & 0.2153 \\ ? & ? & -0.3672 \end{bmatrix}$$

With five unknowns, the cross-product shortcut does not apply directly since no column or row is fully known; the remaining entries were solved numerically (in Python) by enforcing the three defining properties of a rotation matrix: $\|\mathbf{c}_i\|_2 = 1$, $\mathbf{c}_i \cdot \mathbf{c}_j = 0$ for $i \neq j$, and $\det = 1$. These constraints have a sign ambiguity, so two completions are admissible:

$$\mathbf{R} = \begin{bmatrix} -0.4094 & 0.1166 & -0.9049 \\ -0.2039 & 0.9550 & 0.2153 \\ 0.8893 & 0.2727 & -0.3672 \end{bmatrix} \quad \text{or} \quad \mathbf{R}' = \begin{bmatrix} 0.4094 & 0.1166 & 0.9049 \\ -0.2039 & -0.9550 & 0.2153 \\ 0.8893 & -0.2727 & -0.3672 \end{bmatrix}$$

$\mathbf{R}$ above is one of two admissible completions (the other, $\mathbf{R}'$ above, corresponds to the opposite sign of a_{22} , which then flips a_{11} , a_{13} , and a_{32}), and both are verified correct by satisfying unit column norms, mutual orthogonality, and $\mathbf{c}_1 \times \mathbf{c}_2 = \mathbf{c}_3$ (so $\det = +1$) while exactly reproducing the four given entries, meaning those entries alone do not determine a unique rotation matrix.

Appendix A: Python Code

Problem 1: Path and Frame Plot

Listing 1: Problem 1.py: generates the scaled path drawing in Fig. 1.

```

1 import os
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from matplotlib.lines import Line2D
5
6 SCRIPT_DIR = os.path.dirname(os.path.abspath(__file__))
7 FIGURES_DIR = os.path.join(SCRIPT_DIR, '..', 'Latex', 'Figures')
8 FIGURE_PATH = os.path.join(FIGURES_DIR, 'Problem1A.png')
9
10
11 def rot(deg):
12     t = np.radians(deg)
13     return np.array([[np.cos(t), -np.sin(t)], [np.sin(t), np.cos(t)]])
14
15
16 def draw_frame(ax, origin, phi, color, length=0.3, label=None, label_angle
    =0.0):
17     R = rot(phi)
18     x_axis = R[:, 0] * length
19     y_axis = R[:, 1] * length
20     x_patch = ax.arrow(*origin, *x_axis, head_width=0.05, color=color,
        length_includes_head=True, zorder=3)
21     x_patch.set_linestyle('solid')
22     y_patch = ax.arrow(*origin, *y_axis, head_width=0.05, color=color,
        length_includes_head=True, zorder=3)
23     y_patch.set_linestyle('dashed')
24     if label:
25         t = np.radians(label_angle)
26         xytext = (18 * np.cos(t), 18 * np.sin(t))
27         ax.annotate(label, origin, textcoords='offset points', xytext=
            xytext, fontsize=11,
28                     fontweight='bold', color=color, ha='center', va='
                center')
29
30
31 steps = [
32     [('translate', 1.0), ('rotate', -45)],
33     [('translate', 0.5), ('rotate', 30)],
34     [('translate', -2.0)],
35 ]
36
37 pos = np.array([0.0, 0.0])
38 phi = 0.0
39 path = [pos.copy()]
40 frames = [(pos.copy(), phi)]
41
42 for step in steps:
43     for kind, val in step:

```

```

44     if kind == 'rotate':
45         phi += val
46     else:
47         pos = pos + val * rot(phi)[: , 1]
48         path.append(pos.copy())
49     frames.append((pos.copy(), phi))
50
51 path = np.array(path)
52 final_pos, final_phi = frames[-1]
53 obj_pos = final_pos + 1.5 * rot(final_phi)[: , 0]
54
55 colors = plt.get_cmap('tab10').colors
56
57 fig, ax = plt.subplots(figsize=(8, 8))
58 ax.plot(path[: , 0], path[: , 1], 'k-o', linewidth=2, markersize=6, zorder
59         =2)
60 ax.plot(*obj_pos, 'b^', markersize=10, zorder=3)
61
62 for i, (p, ang) in enumerate(frames):
63     draw_frame(ax, p, ang, colors[i % len(colors)], label=str(i),
64               label_angle=-90 + 65 * i)
65
66 obj_color = colors[len(frames) % len(colors)]
67 draw_frame(ax, obj_pos, final_phi, obj_color, label='obj', label_angle
68           =-90)
69
70 all_pts = np.vstack([path, obj_pos])
71 xmin, ymin = np.floor(all_pts.min(axis=0) * 2 - 1) / 2
72 xmax, ymax = np.ceil(all_pts.max(axis=0) * 2 + 1) / 2
73
74 ax.set_xticks(np.arange(xmin, xmax + 0.01, 0.5))
75 ax.set_yticks(np.arange(ymin, ymax + 0.01, 0.5))
76 ax.set_xlim(xmin, xmax)
77 ax.set_ylim(ymin, ymax)
78 ax.set_aspect('equal')
79 ax.grid(True)
80 ax.set_xlabel('x (m)')
81 ax.set_ylabel('y (m)')
82 ax.set_title('Mobile Robot Path')
83
84 legend_elements = [
85     Line2D([0], [0], color='k', marker='o', label='Robot path'),
86     Line2D([0], [0], color='b', marker='^', linestyle='None', label='
87     Object'),
88     Line2D([0], [0], color='gray', linestyle='solid', label='x axis'),
89     Line2D([0], [0], color='gray', linestyle='dashed', label='y axis'),
90 ]
91 ax.legend(handles=legend_elements, loc='upper right')
92
93 os.makedirs(FIGURES_DIR, exist_ok=True)
94 plt.savefig(FIGURE_PATH, dpi=200)
95 plt.show()

```

Problem 2: Manipulator Plot

Listing 2: Problem 2.py: generates the scaled manipulator drawing in Fig. 2.

```

1 import os
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from matplotlib.lines import Line2D
5
6 SCRIPT_DIR = os.path.dirname(os.path.abspath(__file__))
7 FIGURES_DIR = os.path.join(SCRIPT_DIR, '..', 'Latex', 'Figures')
8 FIGURE_PATH = os.path.join(FIGURES_DIR, 'Problem2A.png')
9
10
11 def rot(deg):
12     t = np.radians(deg)
13     return np.array([[np.cos(t), -np.sin(t)], [np.sin(t), np.cos(t)]])
14
15
16 def draw_frame(ax, origin, phi, color, length=0.3, label=None, label_angle
17 =0.0):
18     R = rot(phi)
19     x_axis = R[:, 0] * length
20     y_axis = R[:, 1] * length
21     x_patch = ax.arrow(*origin, *x_axis, head_width=0.05, color=color,
22     length_includes_head=True, zorder=3)
23     x_patch.set_linestyle('solid')
24     y_patch = ax.arrow(*origin, *y_axis, head_width=0.05, color=color,
25     length_includes_head=True, zorder=3)
26     y_patch.set_linestyle('dashed')
27     if label:
28         t = np.radians(label_angle)
29         xytext = (26 * np.cos(t), 26 * np.sin(t))
30         ax.annotate(label, origin, textcoords='offset points', xytext=
31         xytext, fontsize=11,
32         fontweight='bold', color=color, ha='center', va='
33         center')
34
35
36 theta0 = 45.0    # deg, joint 0 (rotary)
37 a1 = 0.5         # m, joint 1 (prismatic)
38 theta2 = 45.0    # deg, joint 2 (rotary)
39
40 # Each step is (fixed/variable rotation about z, then translation along
41 # the resulting x axis)
42 steps = [
43     [('rotate', theta0), ('translate', 1.0)],    # joint 0 + link 1
44     [('rotate', -45.0), ('translate', a1)],    # fixed -45 offset +
45     # prismatic link 2
46     [('rotate', theta2), ('translate', 1.0)],    # joint 2 + link 3
47 ]
48
49 pos = np.array([0.0, 0.0])
50 phi = 0.0

```

```

44 path = [pos.copy()]
45 frames = [(pos.copy(), phi)]
46
47 for step in steps:
48     for kind, val in step:
49         if kind == 'rotate':
50             phi += val
51         else:
52             pos = pos + val * rot(phi)[: , 0]
53             path.append(pos.copy())
54         frames.append((pos.copy(), phi))
55
56 path = np.array(path)
57
58 colors = plt.get_cmap('tab10').colors
59
60 fig, ax = plt.subplots(figsize=(8, 8))
61 ax.plot(path[:, 0], path[:, 1], 'k-', linewidth=2, zorder=2)
62 ax.plot(*path[0], 'ko', markersize=8, zorder=3)           # 00 (rotary
63     base)
64 ax.plot(*path[1], 'ks', markersize=8, zorder=3)           # 01 (prismatic
65     joint)
66 ax.plot(*path[2], 'ko', markersize=8, zorder=3)           # 02 (rotary
67     joint)
68 ax.plot(*path[3], 'k^', markersize=10, zorder=3)          # 03 (end
69     effector)
70
71 frame_labels = ['0', '1', '2', '3']
72 for i, (p, ang) in enumerate(frames):
73     draw_frame(ax, p, ang, colors[i % len(colors)], label=frame_labels[i],
74         label_angle=ang + 225)
75
76 xmin, ymin = np.floor(path.min(axis=0) * 2 - 1) / 2
77 xmax, ymax = np.ceil(path.max(axis=0) * 2 + 1) / 2
78
79 ax.set_xticks(np.arange(xmin, xmax + 0.01, 0.5))
80 ax.set_yticks(np.arange(ymin, ymax + 0.01, 0.5))
81 ax.set_xlim(xmin, xmax)
82 ax.set_ylim(ymin, ymax)
83 ax.set_aspect('equal')
84 ax.grid(True)
85 ax.set_xlabel('x (m)')
86 ax.set_ylabel('y (m)')
87 ax.set_title(r'Three-Link RPR Manipulator ( $\theta_0=\pi/4$ ,  $a_1=0.5\text{m}$ ,  $\theta_2=\pi/4$ )')
88
89 legend_elements = [
90     Line2D([0], [0], color='k', marker='o', linestyle='None', label='
91         Rotary joint'),
92     Line2D([0], [0], color='k', marker='s', linestyle='None', label='
93         Prismatic joint'),
94     Line2D([0], [0], color='k', marker='^', linestyle='None', label='End
95         effector'),
96     Line2D([0], [0], color='gray', linestyle='solid', label='x axis'),

```

```
89     Line2D([0], [0], color='gray', linestyle='dashed', label='y axis'),  
90 ]  
91 ax.legend(handles=legend_elements, loc='upper left')  
92  
93 os.makedirs(FIGURES_DIR, exist_ok=True)  
94 plt.savefig(FIGURE_PATH, dpi=200)  
95 plt.show()
```

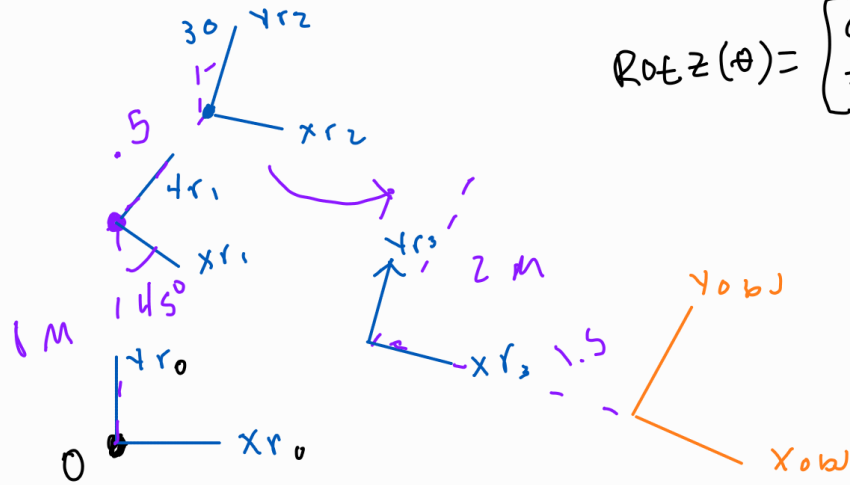
Appendix B: Hand Calculations

Corrections made when typing up the hand calcs below:

1. **Problem 1(b), page 18:** the second rotation was written as $\text{Rot}_z(15^\circ)$, but the problem statement specifies a 30° counter-clockwise turn, and the numeric matrix used in part (c) on the same page is built from $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\sin 30^\circ = \frac{1}{2}$, so $\text{Rot}_z(30^\circ)$ is what was actually (correctly) carried through the rest of the problem. Typed up as $\text{Rot}_z(30^\circ)$.
2. **Problem 1(e), page 19:** the boxed ${}^{\text{obj}}T_0$ had the rotation block negated ($-{}^0R_{\text{obj}}^T$ used for the whole matrix rather than just the translation term), giving $\begin{bmatrix} -0.9659 & 0.2588 \\ -0.2588 & -0.9659 \end{bmatrix}$ and a translation of $(-1.4556, 0.6105)$. Since $\{\text{obj}\}$ shares its orientation with $\{\text{final}\}$, ${}^0R_{\text{obj}} = R(-15^\circ)$, so ${}^{\text{obj}}R_0 = {}^0R_{\text{obj}}^T = R(15^\circ) = \begin{bmatrix} 0.9659 & -0.2588 \\ 0.2588 & 0.9659 \end{bmatrix}$ (positive diagonal), with translation $-{}^0R_{\text{obj}}^T {}^0d_{0 \rightarrow \text{obj}} \approx (-1.4910, 0.6010)$. Typed up with the corrected values.
3. **Problem 2(b), page 20:** the boxed 0T_1 had $s\theta_0$ in the (2,2) entry instead of $c\theta_0$, a copy slip, since the correct $\cos \theta_0$ pattern is shown directly above it (in 0R_1 on the same page) and is used consistently in the part (e) derivation on page 23. Typed up with $c\theta_0$.

Problem 1

a)



$$\text{Rot}_Z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

b)

$${}^0T_{\text{Final}} = \text{Trans}\begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{Rot}(-45) \\ \text{Trans}\begin{pmatrix} 0 \\ .5 \end{pmatrix} \text{Rot}(15) \\ \text{Trans}\begin{pmatrix} 0 \\ -2 \end{pmatrix}$$

c)

$${}^0T_{\text{Final}} = {}^0T_1 {}^1T_2 {}^2T_3$$

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & .5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_{\text{Final}} = \begin{bmatrix} .9659 & .2588 & -.1641 \\ -.2588 & .9659 & -.5782 \\ 0 & 0 & 1 \end{bmatrix}$$

$$d) {}^F T_{obj} = \begin{bmatrix} {}^F R_{obj} & {}^F d_{F \rightarrow obj} \\ \vec{0}^T & 1 \end{bmatrix}$$

$$\text{Final } T_{obj} = \begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

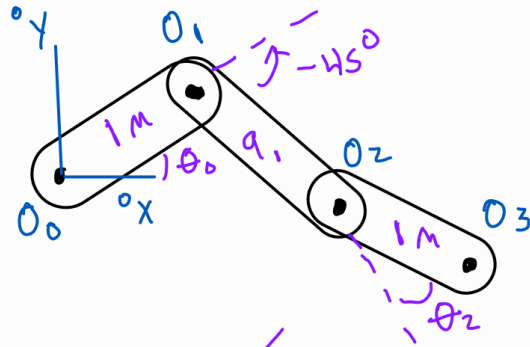
$$e) {}^0 T_{obj} = {}^0 T_{final} \cdot \text{Final } T_{obj}$$

$$= \begin{bmatrix} .9659 & .2588 & 1.2848 \\ -.2588 & .9659 & -.9665 \\ 0 & 0 & 1 \end{bmatrix}$$

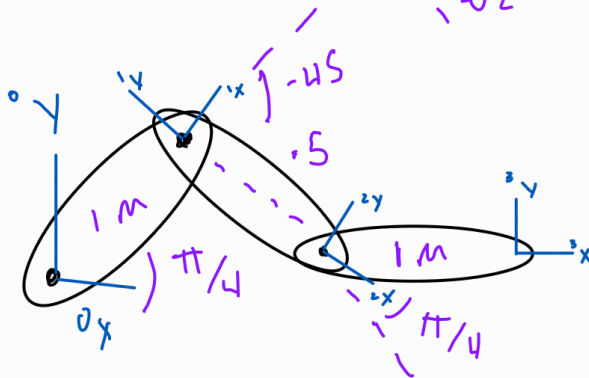
$${}^{obj} T_0 = {}^0 T_{obj}^{-1} = \begin{bmatrix} {}^0 R_{obj}^{-1} & -{}^0 R_{obj}^{-1} {}^0 d_{0 \rightarrow obj} \\ \vec{0} & 1 \end{bmatrix}$$

$${}^{obj} T_0 = \begin{bmatrix} -.9659 & .2588 & -1.4556 \\ -.2588 & -.9659 & .6105 \\ 0 & 0 & 1 \end{bmatrix}$$

Problem 2



a)



$${}^0R_1 = \begin{bmatrix} c\theta & s\theta \\ -s\theta & c\theta \end{bmatrix} \therefore$$

b)

$${}^0T_1 = \begin{bmatrix} {}^0R_1 & {}^0d_{01} \\ \underline{\vec{0}} & 1 \end{bmatrix}$$

$${}^0R_1 = \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix}$$

$${}^1d_{01} = 1 \hat{x} \quad {}^0d_{01} = 1 c\theta \hat{x} + 1 s\theta \hat{y}$$

$$\therefore {}^0T_1 = \begin{bmatrix} c\theta_0 & -s\theta_0 & c\theta_0 \\ s\theta_0 & s\theta_0 & s\theta_0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$c) \quad {}^1T_2 = \begin{bmatrix} {}^1R_2 & {}^1d_{12} \\ \vec{0} & 1 \end{bmatrix}$$

$${}^2R_1 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{bmatrix} \therefore {}^1R_2 = \begin{bmatrix} \cos\theta_1 & \sin\theta_1 \\ -\sin\theta_1 & \cos\theta_1 \end{bmatrix}$$

$$\theta_1 = -45^\circ \therefore {}^1R_2 = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

$${}^2d_{12} = a_1 \hat{x} \quad {}^1d_{12} = a_1 \cos\theta_1 \hat{x} - a_1 \sin\theta_1 \hat{y}$$

$$\therefore$$

$${}^1T_2 = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 & a_1 \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 & -a_1 \sqrt{2}/2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$d) \quad {}^2T_3 = \begin{bmatrix} {}^2R_3 & {}^2d_{23} \\ \underline{\vec{0}} & 1 \end{bmatrix}$$

$${}^2R_3 = \begin{bmatrix} \cos\theta_2 & \sin\theta_2 \\ -\sin\theta_2 & \cos\theta_2 \end{bmatrix} \quad {}^3R_2 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 \\ \sin\theta_2 & \cos\theta_2 \end{bmatrix}$$

$${}^3d_{23} = 1 \quad {}^3\hat{X} \quad {}^2d_{23} = \cos\theta_2 \hat{X} + \sin\theta_2 \hat{Y}$$

$$\therefore {}^2T_3 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & \cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & \sin\theta_2 \\ 0 & 0 & 1 \end{bmatrix}$$

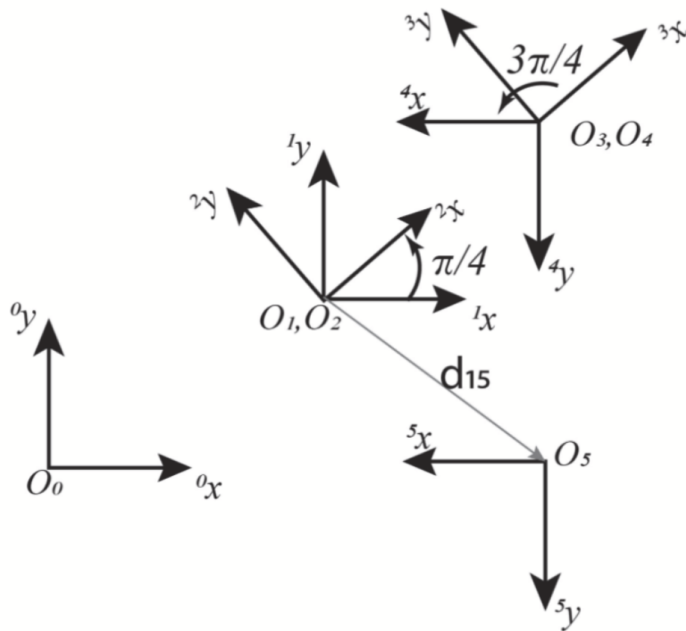
$$E) \quad {}^0T_3 = {}^0T_1 \cdot {}^1T_2 \cdot {}^2T_3$$

$$\begin{bmatrix} \cos\theta_0 & -\sin\theta_0 & \cos\theta_0 \\ \sin\theta_0 & \sin\theta_0 & \sin\theta_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 & a_1\sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 & -a_1\sqrt{2}/2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & \cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & \sin\theta_2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\sqrt{2}}{2} \cos\theta_2 + \frac{\sqrt{2}}{2} \sin\theta_2 & \frac{\sqrt{2}}{2} \cos\theta_2 - \frac{\sqrt{2}}{2} \sin\theta_2 & \frac{\sqrt{2}}{2} \cos\theta_2 + \frac{\sqrt{2}}{2} \sin\theta_2 + a_1 \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \cos\theta_2 + \frac{\sqrt{2}}{2} \sin\theta_2 & \frac{\sqrt{2}}{2} \cos\theta_2 + \frac{\sqrt{2}}{2} \sin\theta_2 & -\frac{\sqrt{2}}{2} \cos\theta_2 + \frac{\sqrt{2}}{2} \sin\theta_2 - a_1 \frac{\sqrt{2}}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

$$OT_3 = \sqrt{\begin{matrix} c(\theta_0 - 45^\circ + \theta_2) & -5(\theta_0 - 45^\circ + \theta_2) & c\theta_0 + a_1 c(\theta_0 - 45^\circ) + c(\theta_0 - 45^\circ + \theta_2) \\ c(\theta_0 - 45^\circ + \theta_2) & 0 & 5\theta_0 + a_1 5(\theta_0 - 45^\circ) + 5(\theta_0 - 45^\circ + \theta_2) \\ 0 & 0 & 1 \end{matrix}}$$

Problem 3



$${}^0d_{01} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad {}^1d_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad {}^2d_{23} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$${}^3d_{34} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad {}^4d_{45} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

a) ${}^0T_1 = \begin{bmatrix} {}^0R_1 & {}^0d_{01} \\ \vec{0} & 1 \end{bmatrix} \quad {}^0R_1 = \underline{\underline{\begin{bmatrix} \cos(\pi/4) & -\sin(\pi/4) \\ \sin(\pi/4) & \cos(\pi/4) \end{bmatrix}}}$

$${}^0T_1 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$b) \quad {}^1T_2 = \begin{bmatrix} {}^1R_2 & {}^1d_{12} \\ \vec{0}^T & 1 \end{bmatrix} \quad {}^1R_2 = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\theta = \pi/4$$

$${}^1T_2 = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad {}^1R_2 = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

$$c) \quad {}^2T_3 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sin, \cos(3\pi/4) = \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}$$

$$d) \quad {}^3T_4 = \begin{bmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$e) \quad {}^4T_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$F) \quad {}^0T_5 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5$$

$${}^0T_5 = \begin{bmatrix} -1 & 0 & 3.41 \\ 0 & -1 & -.59 \\ 0 & 0 & 1 \end{bmatrix}$$

$$G) \quad {}^0\underline{d}_{05} = \begin{bmatrix} 3.41 \\ -.59 \end{bmatrix}$$

$$H) \quad {}^0\underline{R}_5 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$p = \begin{bmatrix} d_{ij} \\ 1 \end{bmatrix}$$

$$i) \quad {}^s p_{s1} = {}^0 T_s^{-1} \cdot \begin{bmatrix} {}^0 d_{01} \\ 1 \end{bmatrix}$$

$$I) \quad {}^s d_{1s} = -{}^s d_{s1}$$

$$\begin{bmatrix} {}^1 d_{1s} \\ 1 \end{bmatrix} = {}^1 T_s \begin{bmatrix} {}^s d_{1s} \\ 1 \end{bmatrix}$$

$$II) \quad {}^0 T_s^{-1} = \begin{bmatrix} {}^0 R_s^{-1} & -{}^0 R_s^{-1} \cdot \underline{d_{0s}} \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 3.41 \\ 0 & -1 & -1.59 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^s d_{s1} = \begin{bmatrix} 1.41 \\ -1.59 \end{bmatrix}$$

$$\therefore {}^s d_{1s} = \begin{bmatrix} -1.41 \\ 1.59 \end{bmatrix}$$

Problem 4

a) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\det = 0 \cdot (\text{sum stuff})$

$$1(1 - 0)$$

$$0(\text{sum stuff})$$

$$= -1$$

$\therefore$ NOT a rotation

b) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ each col norm = 1
 $C_i \cdot C_j = 0$
 $\det(B) = 1$

yes rotation

c) $\|C_i\|_2 = 1, C_i \cdot C_j = 0, \det(C) = 1$
yes rotation

d) $\|C_i\|_2 \neq 1 \therefore$ Not rotation

Problem 5

a)
$$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad C_1 \times C_2 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

b)
$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad C_1 \times C_2 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

c)
$$\begin{bmatrix} .3698 & .3352 & .8665 \\ -.9276 & .1858 & .324 \\ -.05 & -.9236 & .3797 \end{bmatrix} \quad C_2 = C_3 \times C_1 = \begin{bmatrix} .3352 \\ .1858 \\ -.9235 \end{bmatrix}$$

d) used python to solve using $\|C\|_2 = 1$

$$\begin{bmatrix} -.4049 & .1166 & -.9049 \\ -.2039 & .9550 & .2153 \\ .8893 & .2727 & -.3673 \end{bmatrix}$$

$C_1 \cdot C_2 = 0$
det = 1
properties